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***Abstract*—Click-Through Rate (CTR) is a crucial indicator of the relevance of a search result for the user. High search visibility coupled with low CTRs are opportunity pages for optimization. By applying a machine learning approach to CTR opportunity scoring with the FlyRank ML Internship Warehouse data set, this paper provides a novel insight into the challenge.This paper introduces a novel perspective on the challenge by applying a machine learning approach to CTR opportunity scoring using the data set found in the FlyRank ML Internship Warehouse. The search performance data is aggregated to the page level with DuckDB, then CTR is predicted from regression modeling. The resulting rankings give evidence-based information for what pages should be prioritized for optimisation.**

***Index Terms*—Learning Outcomes: Machine Learning, Search Intelligence, Click-Through Rate, Regression, Search Optimization, Data Analytics**

I. Introduction

There's a ton of performance data being generated by search engines that will help to inform decisions about content optimization. As websites get bigger, however, it becomes more difficult to manually find pages that need optimizing.

This project implements a machine learning pipeline that can estimate the probability of a page to have a good CTR and thus high optimization potential based on historical search performance metrics. The workflow enables ranking of pages based on their expectations for improvement and aid in decision making.

II. Problem Statement

This study aims to find pages that have high visibility in search results and low CTR. These pages might need to be optimized for title, meta description or content.

The research question is:

Are there any pages that show meaningful opportunities to improve CTR if analyzed by machine learning models, based on past search performance?

III. Dataset

This study uses the FlyRank ML Internship Warehouse dataset.

The analysis is based on the `fact_content_daily_performance` table.

Using DuckDB, data from daily observations were aggregated into one row per content page.

The following features were used:

- Impressions
- Clicks
- Average Search Position
- Pageviews
- Sessions
- Scroll Events

Click-Through Rate (CTR):

$$CTR = \frac{\text{Clicks}}{\text{Impressions}} \tag{1}$$

IV. Methodology

A. Data Processing

The Hugging Face warehouse was queried directly and without downloading the entire dataset using DuckDB.

Statistics were computed on a daily basis and aggregated to the page level to be suitable for machine learning.

B. Feature Engineering

The following predictor variables were chosen:

- Impressions
- Average Position
- Pageviews
- Sessions
- Scroll Events

CTR was used as the prediction target.

C. Model Development

The data set was split into two parts: training data (80

Two regression models were tested:

- Linear Regression
- Random Forest Regressor

D. Evaluation Metrics

Performance of the models was assessed based on:

- Mean Absolute Error (MAE) Mean Squared Error (MSE)
- R^2 Score

V. Results

The model performance is summarized in the following Table I.

TABLE I
Regression Model Performance

Model	MAE	MSE	R^2
Linear Regression	0.00906	0.00118	0.02475
Random Forest	0.00850	0.00148	-0.21712

The Random Forest model has significantly outperformed the baseline Linear Regression model.

VI. Exploratory Data Analysis

Data samples of CTR and search performance metrics were analysed to get an understanding of their distribution.

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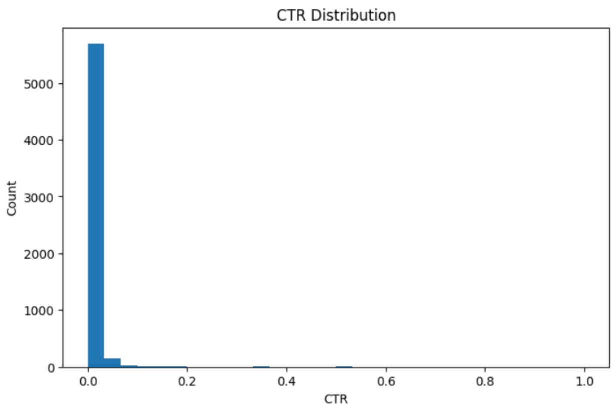

Fig. 1. Distribution of Click-Through Rate

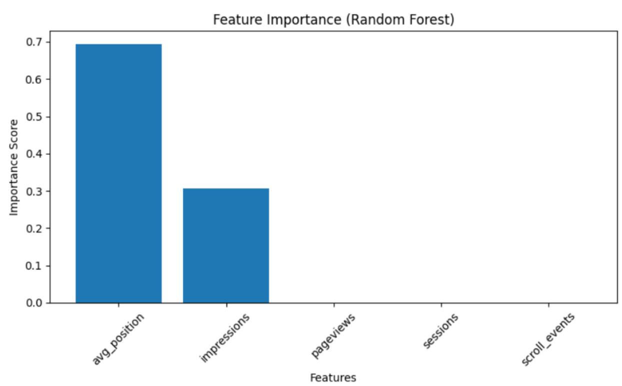

Fig. 2. Feature Importance

VII. Feature Importance

The feature importance analysis revealed the key variables that are most important in the prediction of the CTR.

The results show that the impressions, average search position and user engagement metrics have the highest impact on the predicted CTR.

VIII. Ranked Recommendations

Predicted CTR values were used to generate an opportunity score to guide pages for optimization.

The top ranked pages are suggested for:

- Enhancing the titles of pages Optimizing Meta descriptions
- Updating content
- Enhancing the experience for users How to track future performance

IX. Limitations

There are some drawbacks to this work.

The analysis is carried out using observational data from the past. The model does not identify causal relationships but associations. Only aggregated page level metrics were used. Not included: Additional content quality features.

Thus, it is recommended that the results be used as a basis for making recommendations for decision support.

X. Conclusion

This study shows how to use search performance data to build a machine learning framework for CTR opportunity scoring.

The result was that DuckDB was able to process large search data efficiently, and the pages identified by regression models might need optimization.

The results deliver ranked recommendations, creating a repeatable process to support search optimization decisions.

Acknowledgment

Constructed using the dataset ‘Built on the FlyRank ML Internship’.

References

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